Supplementary Material

We provide additional information to the main content, including an ablation study, some failure cases, and further implementation details. We include more results, in terms of videos, on the (local) supplementary HTML page. The HTML page is fully self-contained with no external links.

6 Ablation study

Regularization strength for deviation. We ablate the impact of the regularizer strength in Equation (11) for how much the box deviates from user intent. We show our results with Trailblazer Ma et al. (2024b) in Figure 11. As shown, stronger regularization leads to bounding boxes that are more faithful to the user input, but their generation results are poor in quality. Too low values create too much deviation. λ_{reg}=0.1 strikes a good balance between user intent and generation quality.

Linear trajectories. We also show our method applied to linear trajectories to simulate a user input. In Figure 12, we use T2V-Turbo Li et al. (2025) with our method and provide a bounding-box that linearly interpolates between a start and an end bounding box. As shown, our method provides enhancement.

Attention map	Attention reshaped	# Edit channels	Edited copy (lower channels)	Layer info
240, 1600, 77	240, 40, 40, 77	120	120, 40, 40, 77	down_blocks.0.0.0
240, 1600, 77	240, 40, 40, 77	120	120, 40, 40, 77	down_blocks.0.1.0
480, 400, 77	480, 20, 20, 77	240	240, 20, 20, 77	down_blocks.1.0.0
480, 400, 77	480, 20, 20, 77	240	240, 20, 20, 77	down_blocks.1.1.0
960, 100, 77	960, 10, 10, 77	480	480, 10, 10, 77	down_blocks.2.0.0
960, 100, 77	960, 10, 10, 77	480	480, 10, 10, 77	down_blocks.2.1.0
960, 25, 77	960, 5, 5, 77	480	480, 5, 5, 77	mid_block.0.0
960, 100, 77	960, 10, 10, 77	480	480, 10, 10, 77	up_blocks.1.0.0
960, 100, 77	960, 10, 10, 77	480	480, 10, 10, 77	up_blocks.1.1.0
960, 100, 77	960, 10, 10, 77	480	480, 10, 10, 77	up_blocks.1.2.0
480, 400, 77	480, 20, 20, 77	240	240, 20, 20, 77	up_blocks.2.0.0
480, 400, 77	480, 20, 20, 77	240	240, 20, 20, 77	up_blocks.2.1.0
480, 400, 77	480, 20, 20, 77	240	240, 20, 20, 77	up_blocks.2.2.0
240, 1600, 77	240, 40, 40, 77	120	120, 40, 40, 77	up_blocks.3.0.0
240, 1600, 77	240, 40, 40, 77	120	120, 40, 40, 77	up_blocks.3.1.0
240, 1600, 77	240, 40, 40, 77	120	120, 40, 40, 77	up_blocks.3.2.0

Table 2: **Edited cross attention maps** – As in Trailblazer Ma et al. (2024b), we edit only the lower channels across all layers. Each layer follows the spatial cross-attention naming where *down_blocks.0.0.0* can be fully written as ***down_blocks.0****.attentions.****0****.transformer_blocks.****0****.attn2*.

7 Failure cases

Additionally, 11% and 33% of the users did not have a preference for the quality and trajectory faithfulness questions respectively. This can be attributed to the limitations of the underlying video model both in terms of inability to faithfully generate according to the prompt and at desired locations. Still, with more advanced video models becoming increasingly available Research (2023); Wan et al. (2025), we expect this problem to be mitigated in the future.

Figure 11: **Effect of regularization strength/penalty** – We show an example frame with original bounding boxes as blue and the adjusted boxes as orange. A higher penalty (1.0) allows less movement, while too loose penalty (0.001) causes generation to stray away from user intent. We set the penalty to 0.1 for all our experiments, which strikes a good balance.

8 Further implementation details

Keeping bounding boxes within images. To prevent optimization from making bounding boxes go out of images, we simply clip the optimized bounding box coordinates.

Additional hyperparameters. Besides the hyperparameters noted in the main paper, for others, we follow Trailblazer Ma et al. (2024b) to be comparable. Specifically for Trailblazer Ma et al. (2024b), we use Zeroscope v2 576w as the underlying text-to-video model,[5] and the same DPM Solver[6] for both methods, with a guidance scale of 9.0. We run 40 denoising steps, applying the editing (and our optimization) on the first five steps. For Peekaboo baseline, we run the same number of denoising steps with the default number of frozen steps set to 2. For Freetraj baseline, we use their native resolution (320 × 512) with 50 denoising steps. We also use their DDIM eta of 0, with 6 edit steps, and set the video length to 24 frames, matching the input trajectory.

For T2V-Turbo Li et al. (2025), we use the official implementation, with the default parameters and 16 denoising steps. We apply the edits, both of our method and Trailblazer Ma et al. (2024b) for only the first five steps.

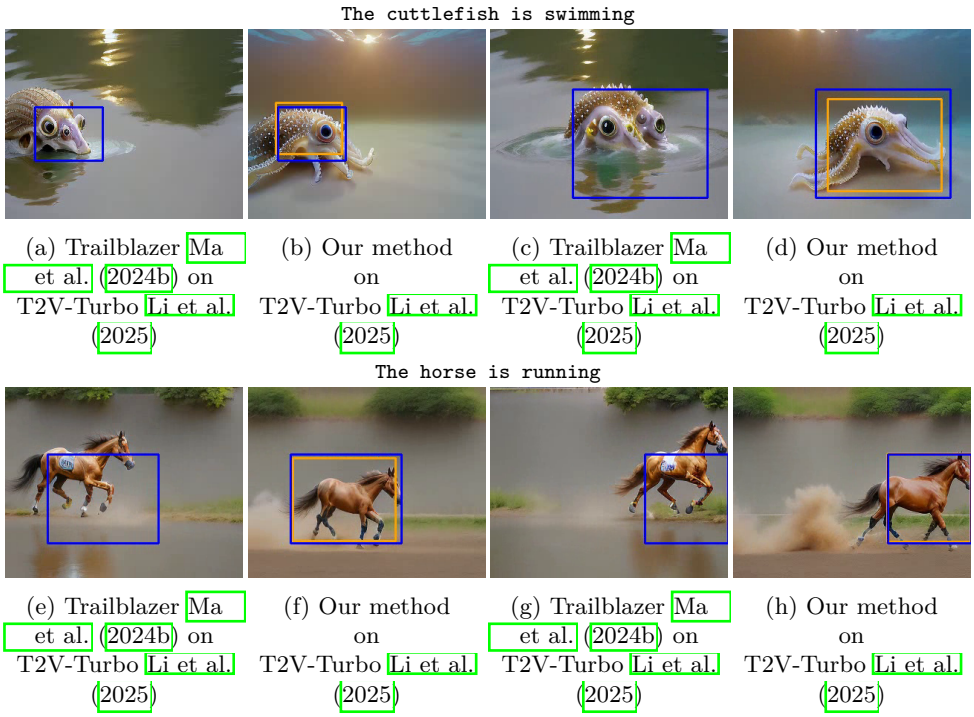
Control metric. For computing the mIoU, we simply choose, among the detected bounding boxes, one that matches most closely to the user control bounding box. Note that this is different from Trailblazer Ma et al. (2024b) which relies on the object detection score. Our strategy takes into account that generated videos are not constrained to generate only a single object, as there may be distractors. We found that Trailblazer Ma et al. (2024b) strategy of using the detection scores does get easily distracted by distractors, thus providing an unfair comparison. Our strategy fixes this.

Model architecture. For Trailblazer Ma et al. (2024b), we use the U-Net architecture of Zeroscope. An initial gaussian noise z_t with shape $1 \times 4 \times \mathrm{F} \times 40 \times 40$ is generated where $\mathrm{F} = 24$ frames. We edit only the spatial attention with a target video resolution of 320×320. The corresponding attention maps are computed from the queries and keys, while the total number of tokens is 77 Radford et al. (2021); each token has a dimension of 1024. The progression of the edited attention maps with their respective dimensions is detailed in Table 2.

For T2V-Turbo Li et al. (2025), the base model utilizes the same U-Net-style architecture as Trailblazer Ma et al. (2024b), hence we edit in the same way as for Trailblazer Ma et al. (2024b). We use PyTorch Paszke et al. (2019) gradient checkpointing in our implementations.

[5] Zeroscope v2 576w model. https://huggingface.co/cerspense/zeroscope_v2_576w

[6] Cheng Lu *et.al.*, DPM-Solver: A Fast ODE Solver for Diffusion Probabilistic Model Sampling in Around 10 Steps, *Advances in Neural Information Processing Systems*, 2022



(a) Trailblazer Ma et al. (2024b) on T2V-Turbo Li et al. (2025) (b) Our method on T2V-Turbo Li et al. (2025) (c) Trailblazer Ma et al. (2024b) on T2V-Turbo Li et al. (2025) (d) Our method on T2V-Turbo Li et al. (2025)

(e) Trailblazer Ma et al. (2024b) on T2V-Turbo Li et al. (2025) (f) Our method on T2V-Turbo Li et al. (2025) (g) Trailblazer Ma et al. (2024b) on T2V-Turbo Li et al. (2025) (h) Our method on T2V-Turbo Li et al. (2025)

Figure 12: **Application to perspective and linear trajectories (top and bottom respectively)** – We show example frames of applying our method to T2V-Turbo Li et al. (2025) with a trajectory that linearly interpolates two bounding boxes (start and end). Ours still delivers improvements, both in terms of generation quality within the bounding box and adherence.

Model ↑	**HPSv2** ↑	**mIOU** ↑
Freetraj Qiu et al. (2024)	0.223	0.34
Our method (T2V-Turbo)	0.263	0.41

Table 3: **Quantitative results with additional baseline using human-preference and control metrics** – Our method outperforms Freetraj, an additional training-free baseline, in terms of human-preferred quality and control accuracy.

Optimization procedure We present our optimization procedure in Algorithm 1 and show how optimization interleaves with the diffusion process.

9 Additional baseline coverage

We additionally compare our method with FreeTraj, a training-free approach for controlled generation, and show that our method achieves higher human-preference scores and improved motion control, as shown in Table 3. Additionally, we show qualitative examples of video frames generated by our method compared to Freetraj in Figure 13. Notably, our method perform significantly better than Freetraj Qiu et al. (2024) in terms of generation quality and adherence to control. Particularly, our method produces better quality, as shown in the *"drosophila melanogaster fruit fly"* example (top row) and adherence to control, as shown in the *"dog"* example (bottom row).

10 Ablations on component design choices

We include comprehensive ablations on key design choices in Tables 4 to 7.

Algorithm 1 Trajectory optimization

Input: Video diffusion model Θ_b, decoder D, initial latents $\boldsymbol{z}$, prompt embedding $\boldsymbol{p}$, denoise steps K, spatial edit steps K_S, temporal edit steps K_T, inner optimization steps O, learning rate $\sqcap$, and trajectory boxes $\mathcal{B}$
for $i \leftarrow 1$ to K **do**
  $r \leftarrow 0$ ▷ inner optimization step counter
  $t \leftarrow \text{TimestepSchedule}(i)$
  while $(i < K_S \vee i < K_T) \wedge (r < O)$ **do**
    /* extract attention maps and noise prediction */
    $(\boldsymbol{A}_S, \boldsymbol{A}_T, \epsilon_t) \leftarrow \Theta_b(\boldsymbol{p}, \boldsymbol{z}, t)$
    /* Apply editing (Eq. 6), compute loss (Eq. 11) and gradients */
    $(\tilde{\boldsymbol{A}}_S, \tilde{\boldsymbol{A}}_T) \leftarrow \text{edit}(\boldsymbol{A}_S, \boldsymbol{A}_T)$ ▷ Eq. 6
    $\mathcal{L} \leftarrow \mathcal{L}(\tilde{\boldsymbol{A}}_S, \tilde{\boldsymbol{A}}_T, \mathcal{B})$ ▷ aggregated over frames and layers; Eq. 11
    $g \leftarrow \nabla_{\mathcal{B}} \mathcal{L}$
    /* Update bounding boxes */
    $\mathcal{B} \leftarrow \text{AdamStep}(\mathcal{B}, g; \sqcap)$
    /* Project boxes to valid boundaries */
    $\mathcal{B} \leftarrow \text{Proj}(\mathcal{B})$
    $r \leftarrow r + 1$
  end while
  /* standard denoise step (no editing) */
  $\epsilon_t \leftarrow \Theta_b(\boldsymbol{p}, \boldsymbol{z}, t)$
  $\boldsymbol{z} \leftarrow \text{SchedulerStep}(\boldsymbol{z}, \epsilon_t, t)$
end for
return $D(\boldsymbol{z})$

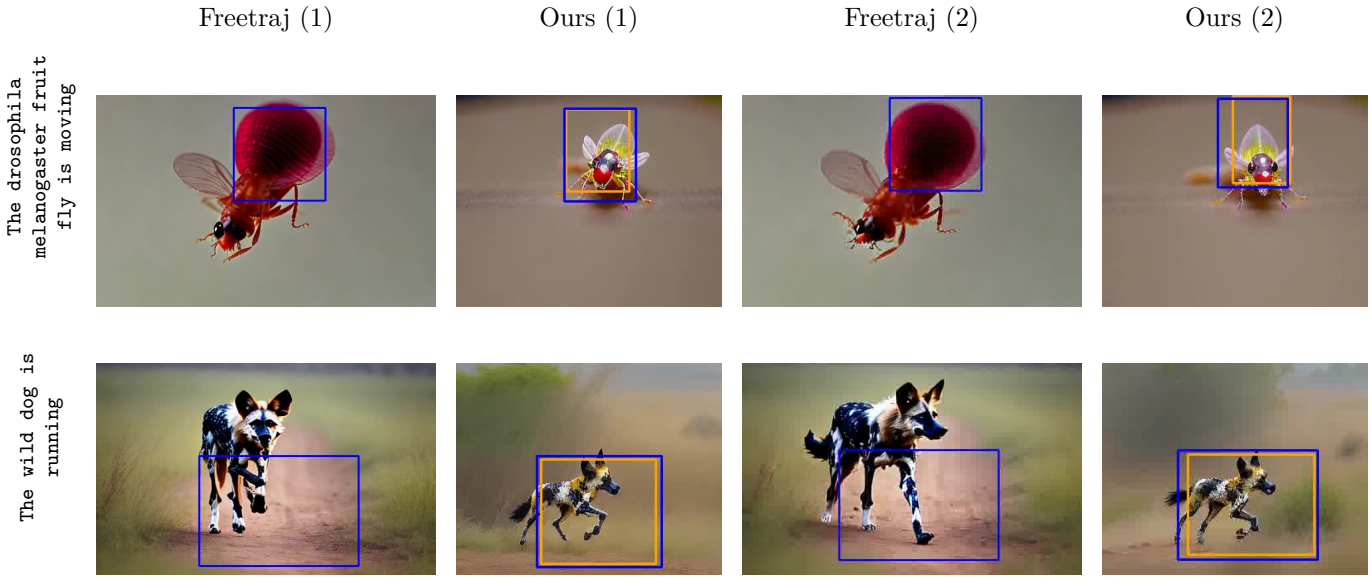


Figure 13: **Qualitative results comparing Freetraj Qiu et al. (2024) and our method on T2V-Turbo Li et al. (2025).** Frames are shown at a common display height (aspect ratio preserved); frame resolutions differ (Freetraj: 320×512, Ours: 256×320). Ours still delivers better generation quality and adherence to control.

Deviation penalty. Table 4 shows that excessive deviation leads to loss of user intent (lower mIoU), while overly restrictive penalties limit quality improvements. A penalty of 0.1 balances control and generation quality.

α	**PickScore** ↑	**HPSv2** ↑	**mIOU** ↑
0	0.210	0.230	0.31
0.001	0.211	0.230	0.32
0.01	0.199	0.228	0.36
0.1	0.194	0.225	0.38
1	0.187	0.221	0.37

Table 4: Ablation results for deviation penalty term $\lambda_{\text{reg}}=\alpha \times \sqrt{A}$

Normalization	**PickScore** ↑	**HPSv2** ↑	**mIOU** ↑
True	0.509	0.225	0.38
False	0.491	0.223	0.38

Table 5: Ablation results for smoothing kernel normalization

$\lambda_{\neg\text{attn}}$	**PickScore** ↑	**HPSv2** ↑	**mIOU** ↑
0	0.228	0.219	0.36
1	0.231	0.221	0.39
10	0.256	0.225	0.38
100	0.285	0.228	0.31

Table 6: Ablation results for background preservation term $\lambda_{\neg\text{attn}}$

λ_{edge}	**PickScore** ↑	**HPSv2** ↑	**mIOU** ↑
0.0001	0.240	0.224	0.40
0.001	0.248	0.226	0.38
0.03	0.243	0.225	0.38
0.1	0.268	0.232	0.36

Table 7: Ablation results for edge strength λ_{edge}

Smoothing kernel normalization. Table 5 demonstrates that normalizing the smoothing kernel to produce consistent peak attention across layers significantly improves PickScore, HPSv2, and mIoU.

Background preservation term. Table 6 confirms that the background attention term is necessary for maintaining scene coherence.

Edge strength. Table 7 shows that very low edge strength behaves similarly to discontinuous masks (e.g., Trailblazer), degrading quality, while overly strong edges reduce control accuracy. Intermediate values (0.001–0.03) provide the best trade-off.

11 Complex patterns and difficult trajectories

We show results for complex patterns and difficult trajectories such as morphing, zig-zag, U-turn, and stationary-to-move in Figures 14 to 16.

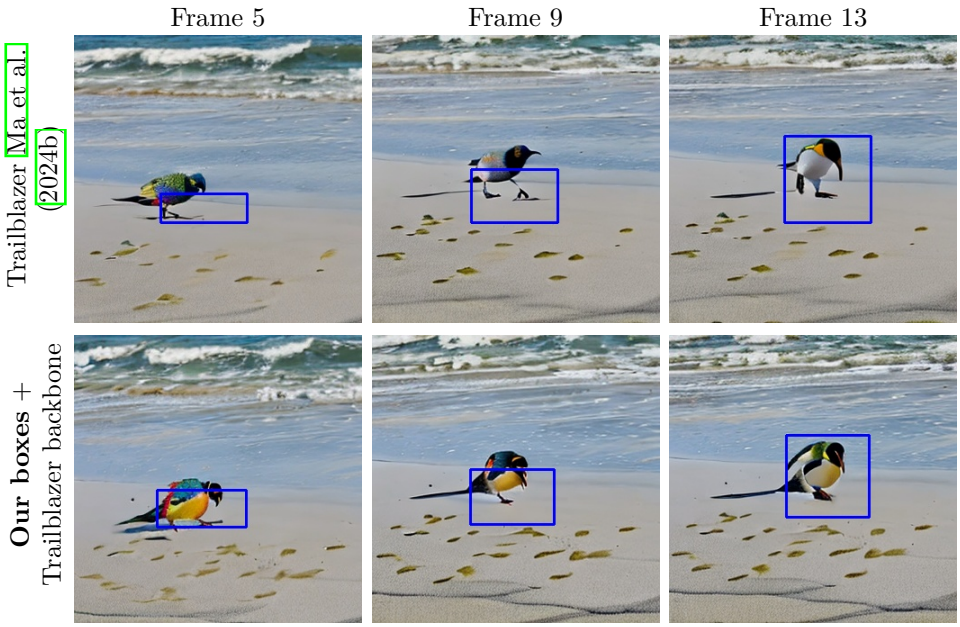


Figure 14: **Additional qualitative results for morphing task (Parrot → Penguin)**– We show transition frames, where our optimized boxes provide benefit for better control of the morphing task.

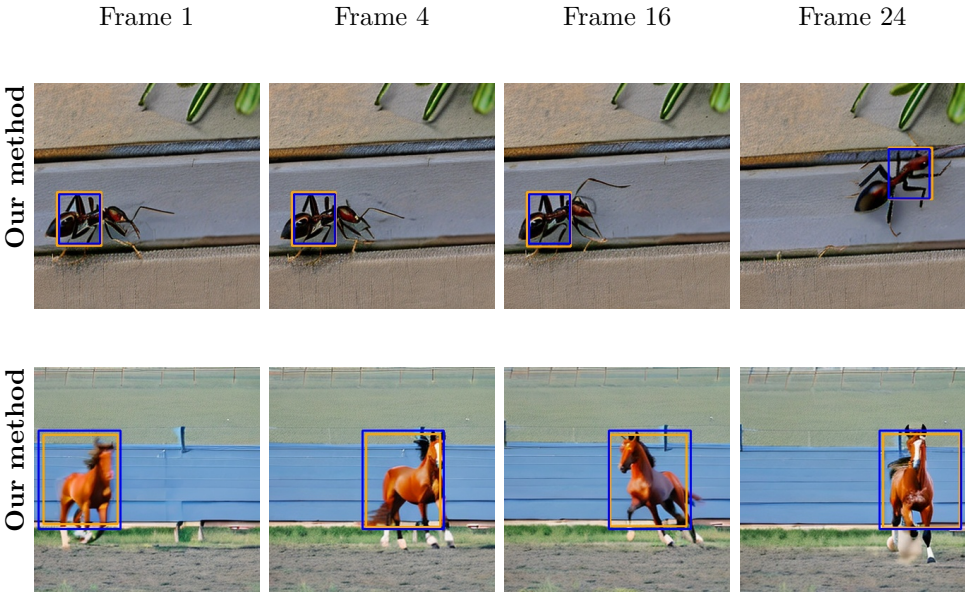


Figure 15: **Additional qualitative results with stationary-to-move and U-turn trajectories** – We show example frames where our method is applied to complex patterns and preserves difficult motions. Best viewed in motion in the supplemental video.

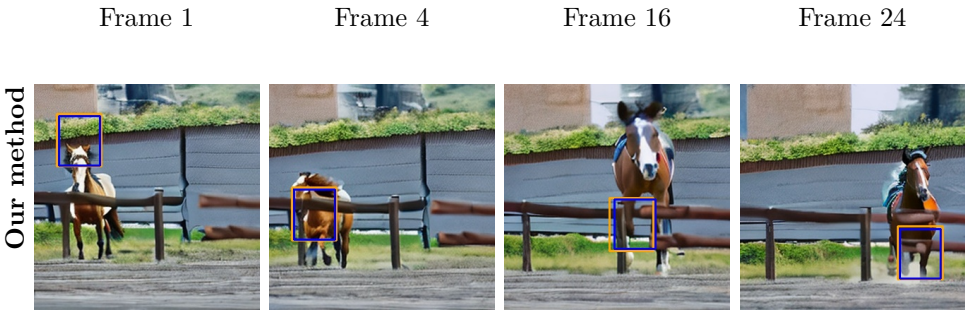


Figure 16: **Extreme scenario case with very small boxes and in a zig-zag trajectory –** We present an extreme scenario with zig-zag motion using very small boxes, different from the object size. Our method compensates and generates motion that follows user intent. Best viewed in motion in the supplemental video.